8.8 Numerical methods

8.8.1 The finite element method in one space dimension

We have introduced in the previous chapter the Galerkin method, that is an approximation method of the abstract variational problem $B(u,v) = Lv$, based on a sequence of finite dimensional subspaces $V_k \subset V$. The finite element method is characterized by a particular definition of the subspaces V_k. We will develop here the main lines of this method applied to the discretization of second order boundary value problems on the interval (x_a, x_b), where in general the space V can be identified with $H^1(x_a, x_b)$.

These preliminary considerations already show that the finite element method relies on completely different theoretical foundations than finite difference schemes. While the latter provide a nodal based approximation, without guarantees on the behavior of the numerical solution between the nodes, the approximation properties of finite element method can be defined and analyzed at any point on the domain. Anyway, for simple problems as the ones that we will address in this section, the two families of methods share some similarities.

For the definition of subspaces V_k we start from a uniform partition of (x_a, x_b), in k subintervals

$$[x_a, x_b] = \bigcup_{i=1}^{k} [x_{i-1}, x_i], \quad x_i = h \cdot i, \quad h = (x_b - x_a)/k.$$

In this context, we say that x_i are the **vertexes** of the partition. The space of **finite elements of degree** r on this partition is defied as the collection of continuous functions whose restriction on each interval $[x_{i-1}, x_i]$ is a polynomial function of degree less or equal to r, namely

$$X_k^r = \{v_k \in C^0(x_a, x_b) \ : \ v_k|_{(x_{i-1}, x_i)} \in \mathbb{P}^r(x_{i-1}, x_i), \text{ for } i = 1, \ldots, k\}$$

where the notation $\mathbb{P}^r(x_{i-1}, x_i)$ stands for the space of polynomial functions of degree less or equal to r on (x_{i-1}, x_i). Clearly, $X_k^r(x_a, x_b)$ is a finite dimensional function space where $\dim(X_k^r) = N_k^r$ is affected by both parameters k and r. Then, the finite element method consists to determine $u_k \in X_k^r(x_a, x_b)$ such that,

$$B(u_k, v_k) = Lv_k, \quad \forall v_k \in V_k.$$

Now, we aim to reformulate the previous discrete problem as a linear system of algebraic equations, that could then be solved by a suitable algorithm. From Chapter 7, we remind that the Galerkin method is equivalent to determine a vector of coefficients, $\mathbf{c} = \{c_j\}_{j=1}^{N_k^r}$, such that $\mathbf{A}_{kr}\mathbf{c} = \mathbf{F}_{kr}$ with $\mathbf{A}_{kr} \in \mathbb{R}^{N_k^r \times N_k^r}$, $\mathbf{F}_{kr} \in \mathbb{R}^{N_k^r}$ where,

$$u_k = \sum_{j=1}^{N_k^r} c_j \psi_j, \quad A_{kr,ij} = B(\psi_j, \psi_i), \quad F_{kr,i} = L\psi_i.$$

According to this equivalent problem formulation, we observe that a suitable basis of the finite element space $X_k^r(x_a, x_b)$ should satisfy the following criteria:

a) The support of each basis element, namely ψ_i, should be small so as to $\text{supp}(\psi_i) \cap \text{supp}(\psi_j) = \emptyset$ for as many indexes as possible. By this way, we a priori know that the matrix coefficient $\mathbf{A}_{kr}$ vanishes because $A_{kr,ij} = a(\psi_j, \psi_i) = 0$. On the one hand, this strategy reduces the computational cost for building A_{kr}. On the other hand, since A_{kr} is likely to be a narrow banded matrix, it will be possible to use efficient algorithms to solve the corresponding linear system.
b) We look for a Lagrangian basis. Namely, provided that each basis element refers ψ_i to a special point $\widehat{x}_i \in (x_a, x_b)$, called **node**, a basis is Lagrangian if $\psi_i(\widehat{x}_j) = \delta_{ij}$, where δ_{ij} is Kronecker's symbol. As a result of that, for any $v_k = \sum_{i=1}^{N_k^r} c_i \psi_i$ the coefficients of the expansion satisfy $c_i = v_k(\widehat{x}_i)$.

Fig. 8.1 shows an example of functions $v_k^1 \in X_k^1(0,1)$ and $v_k^2 \in X_k^2(0,1)$ defined on a uniform partition of $(0,1)$ in $k = 4$ elements, while in Fig. 8.2 we depict some elements of a Lagrangian basis of $X_k^1(0,1)$ (left) and $X_k^2(0,1)$ (right). We observe that in the linear case, nodes and vertexes of the partition coincide. The number of degrees of freedom of the linear finite element space is thus $\dim(X_k^1(0,1)) = N_k^1 = k+1$. Furthermore, each element of the Lagrangian basis is a piecewise linear function with unit value in one particular node and vanishing in all other nodes of the mesh.

For the case of quadratic elements, since a second order polynomial is uniquely defined by fixing it on 3 points, we observe that the set of nodes must be richer than the vertexes of the partition. In particular, it is sufficient to complement the vertexes with the midpoints of each interval in order to obtain a complete set of interpolation nodes that uniquely characterize quadratic finite element functions. As a result of that, we easily conclude that for the one dimensional case $\dim(X_k^2(0,1)) = N_k^2 = 2k+1$ and in general $\dim(X_k^r(0,1)) = N_k^r = rk+1$. Numbering nodes from left to right, Fig. 8.2 (right) shows the

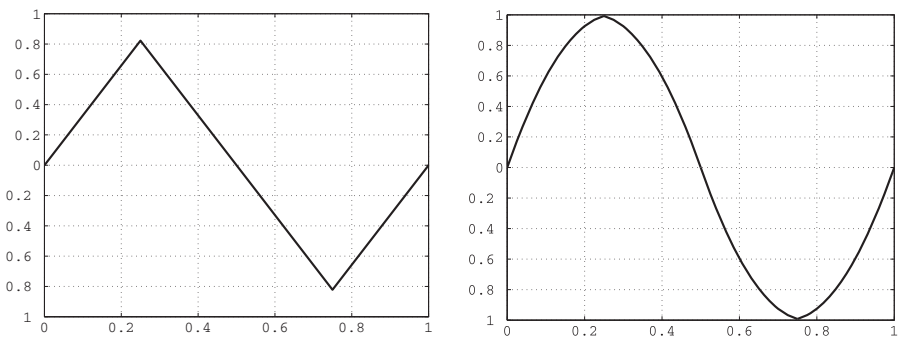

Fig. 8.1. Linear ($r = 1$, left) and quadratic ($r = 2$ right) finite element functions

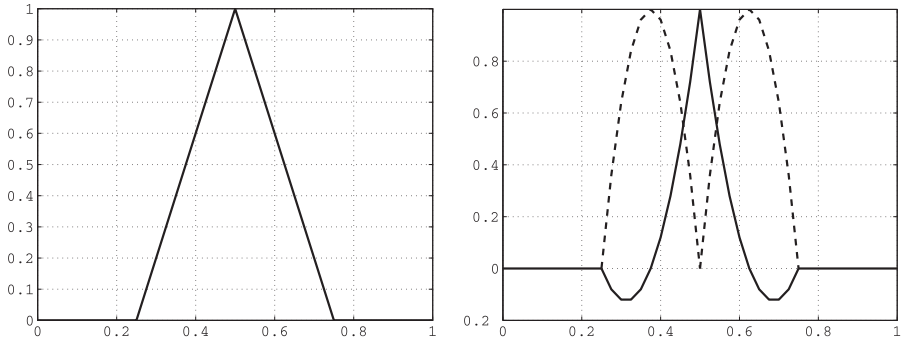

Fig. 8.2. Linear ($r = 1$, left) and quadratic ($r = 2$ right) Lagrangian basis functions for the finite element method

components ψ_4, ψ_5, ψ_6 of the Lagrangian basis for $X_k^2(0,1)$. In particular, ψ_4, ψ_6 are related to the element midpoints and are depicted by a dashed line, while ψ_5 refers to a node coinciding with a vertex and is depicted with continuous line.

The numerical approximation of boundary conditions with the finite element method is completely analogous with their treatment at the level of the abstract variational problem. Homogeneous Dirichlet boundary conditions are enforced into the definition of the discrete space V_k in order to satisfy $V_k \subset V = H_0^1(x_a, x_b)$. To this aim, we define a new discrete space

$$X_{k,0}^r(x_a, x_b) = \{v_k \in X_k^r(x_a, x_b) \ : \ v_k(x_a) = v_k(x_b) = 0\} \subset H_0^1(x_a, x_b)$$

where we have removed the basis components related to nodes $x_0 = x_a$ and $x_{rk} = x_b$ from the original finite element space $X_k^r(x_a, x_b)$. Non homogeneous Dirichlet conditions are reformulated as homogeneous conditions plus a right hand side term by means of their lifting on the entire domain. This technique is straightforwardly extended at the discrete level, provided that a discrete lifting is applied, i.e. a lifting belonging to $X_k^r(x_a, x_b)$. Finally, the discretization of Neumann and Robin conditions do not require any particular care, because they are naturally embedded into the bilinear form.

8.8.2 Error analysis of the finite element method

The Céa Lemma shows that the Galerkin method is convergent provided that the family of approximation spaces V_k becomes dense in $V = H^1(x_a, x_b)$. Namely, given $u \in V$ the solution of the abstract variational problem, we know that

$$\|u - u_k\|_{H^1} \leq \frac{M}{\alpha} \inf_{v_k \in V_k} \|u - v_k\|_{H^1}. \tag{8.79}$$

By consequence, we require that the discrete approximation spaces V_k satisfy the following approximation property

$$\lim_{k\to\infty} \inf_{v_k \in V_k} \|v - v_k\|_{H^1} = 0, \quad \forall v \in H^1(x_a, x_b). \tag{8.80}$$

In this section we analyze how, for a fixed value of the polynomial degree r, the finite element spaces $X_k^r(x_a, x_b)$ satisfy property (8.80) when k increases, or similarly, the partition of the domain (x_a, x_b) is refined by decreasing the characteristic mesh size h. Furthermore, we aim to determine the asymptotic **order of convergence**, that is the largest value of the exponent p such that

$$\lim_{k\to\infty} \|u - u_k\|_{H^1} = \lim_{k\to\infty} C((x_b - x_a)/k)^p = \lim_{h\to 0} Ch^p$$

being C a generic positive constant independent on k, h. The following Theorem[5] is at the basis of the approximation properties of the finite element method.

Theorem 8.7. *For any value of the polynomial degree $r \geq 1$ there exists a positive constant C_r, uniformly independent on k, h such that, for any $v \in H^s(x_a, x_b)$ with $s \geq 2$ it holds*

$$\inf_{v_k \in X_k^r} \|v - v_k\|_{H^1} \leq C_r h^l \|v\|_{H^{l+1}} \quad \textit{where} \quad l = \min[s-1, r], \ h = \frac{x_b - x_a}{k}. \tag{8.81}$$

This shows that the finite element space $X_k^r(x_a, x_b)$ satisfy (8.80) provided that the solution to be approximated is regular enough. More precisely, an *a priori* error estimate for the finite element method is obtained by combining inequalities (8.79) and (8.81), leading to the following result.

Corollary 8.1. *Provided that problem $B(u, v) = Lv$ admits a regular solution $u \in H^s(x_a, x_b)$ with $s \geq 2$, then the finite element approximation of degree $r \geq 1$ satisfies,*

$$\|u - u_k\|_{H^1} \leq C(M, \alpha, r) h^l \|u\|_{H^{l+1}} \tag{8.82}$$

$$\textit{where} \quad l = \min[s-1, r] \quad \textit{and} \quad C(M, \alpha, r) = C_r \frac{M}{\alpha}.$$

Conversely, when no more than $u \in H^1(x_a, x_b)$ can be guaranteed, the asymptotic order of convergence with respect to the characteristic mesh size can be arbitrarily low.

[5] For a proof, we remand the interested reader to Theorem 3.2, Chapter 3 of *Quarteroni* (2009).

8.8.3 The finite element method for the approximation of advection, diffusion, reaction problems

We aim to use the finite element method for the approximation of problem (8.4), that for the sake of clarity is reported below, complemented with homogeneous Dirichlet boundary conditions,

$$-(p(x)u')' + q(x)u' + r(x)u = f(x), \text{ in } (x_a, x_b), \tag{8.83}$$

where $u(x_a) = u(x_b) = 0$, with $p(x), q(x) \in C^1([0,1])$ with $p(x) \geq a_0 > 0$ and $r(x), f(x) \in C^0([0,1])$, whose variational formulation consists to find $u \in H_0^1(x_a, x_b)$ such that

$$\begin{aligned} B(u,v) &= \int_{x_a}^{x_b} (p(x)u'v' + q(x)u'v + r(x)uv)dx = Lv \\ &= \int_{x_a}^{x_b} f(x)vdx, \quad \forall v \in H_0^1(x_a, x_b). \end{aligned}$$

Then, at the discrete level, we look for $u_k \in X_{k,0}^r(x_a, x_b)$ such that $B(u_k, \psi_i) = F\psi_i$ for any $i = 1, \ldots, rk+1$.

We notice that Corollary 8.1 is applicable also to this case and in particular the values of continuity and coercivity constants of the bilinear form, M and α respectively, depend on p, q, r as follows,

$$M = \left(\|p\|_{L^\infty} + C_P\|q\|_{L^\infty} + C_P^2\|r\|_{L^\infty}\right), \quad \alpha = a_0.$$

Accordingly, the a priori error estimate constant $C(M, \alpha, r)$ may become arbitrarily large. By consequence, either the mesh characteristic size is chosen to be be small enough, or the approximation is not satisfactory. In particular, the more the advection terms dominate over the diffusion one, the more the scheme becomes **unstable**. This effect is emphasized in Fig. 8.3 where we show the solution obtained approximating (8.83) and $u(0) = 0,\ u(1) = 1$ by means of linear finite elements. On the left, we show the comparison between approximations obtained when $p = 10^{-2}, q = 1, r = 0$, using uniform meshes characterized by $h = 0.1$ and $h = 0.01$ identified by dashed and continuous lines respectively. On the right, we perform similar numerical tests for a reaction dominated problem with $p = 10^{-2}, q = 0, r = 10$. When the mesh is not refined enough, spurious numerical oscillations clearly appear in both cases.

To stabilize the finite element method, we analyze the scheme at the level of the algebraic nodal equations. To this purpose, we first subdivide the bilinear form $B(u,v)$ into diffusion, transport and reaction terms,

$$B_p(u,v) = \int_{x_a}^{x_b} p(x)u'v'dx, \quad B_q(u,v) = \int_{x_a}^{x_b} q(x)u'vdx,$$
$$B_r(u,v) = \int_{x_a}^{x_b} r(x)uv.$$

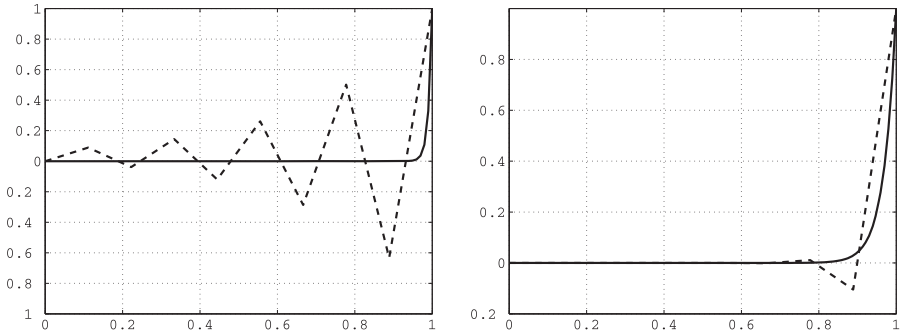

Fig. 8.3. Numerical instability of the finite element method for advection and reaction dominated problems, depicted on the left and right panels respectively. Dashed line corresponds to the approximation obtained using $h = 0.1$, while continuous line corresponds to the numerical solution with $h = 0.01$, which turns out to be very accurate

and in the case of linear finite elements the corresponding algebraic equations become,

$$
\begin{aligned}
B_p(u_k, \psi_i) &= c_{i-1} \int_{x_{i-1}}^{x_i} p(x)\psi'_{i-1}\psi'_i + c_i \int_{x_{i-1}}^{x_{i+1}} p(x)(\psi'_i)^2 + c_{i+1} \int_{x_i}^{x_{i+1}} p(x)\psi'_{i+1}\psi'_i \\
&= -\frac{p}{h}\big(c_{i-1} - 2c_i + c_{i+1}\big), \\
B_q(u_k, \psi_i) &= c_{i-1} \int_{x_{i-1}}^{x_i} q(x)\psi'_{i-1}\psi_i + c_i \int_{x_{i-1}}^{x_{i+1}} q(x)\psi'_i\psi_i + c_{i+1} \int_{x_i}^{x_{i+1}} q(x)\psi'_{i+1}\psi_i \\
&= \frac{q}{2}\big(c_{i+1} - c_{i-1}\big), \\
B_r(u_k, \psi_i) &= c_{i-1} \int_{x_{i-1}}^{x_i} r(x)\psi_{i-1}\psi_i + c_i \int_{x_{i-1}}^{x_{i+1}} r(x)\psi_i^2 + c_{i+1} \int_{x_i}^{x_{i+1}} r(x)\psi_{i+1}\psi_i \\
&= \frac{r}{6}h\big(c_{i+1} + 4c_i + c_{i-1}\big).
\end{aligned}
$$

Since we are using a Lagrangian basis to represent the discrete space, the unknown coefficients c_i coincide with the nodal values of the finite element function, namely $c_i = u_k(x_i) = u_i$. By multiplying by h the above expressions, it emerges that, for uniform partitions, the one dimensional linear finite element approximation of diffusion and transport terms coincide with the corresponding finite difference schemes, while the two families of methods differ in the approximation of reaction terms.

From Chapter 2, we know that for transport dominated problems it is more convenient to discretize qu' using a one sided difference ratio. This observation gives rise to the **upwind** scheme, where the one-sided approximation of the first order derivative is suitably chosen according to the sign of q. Shifting to one-sided schemes does not apply to the finite element framework. However,

we have put into evidence that the upwind scheme is equivalent to a centered difference scheme modified with an **artificial diffusion** term. In particular, for $q > 0$ we easily see that

$$\frac{q\big(u_i - u_{i-1}\big)}{h} = \frac{q\big(u_{i+1} - u_{i-1}\big)}{2h} - \frac{qh}{2}\frac{\big(u_{i+1} - 2u_i + u_{i-1}\big)}{h^2}.$$

Then, for one dimensional and linear finite elements, the correct amount of artificial diffusion that allows us to **stabilize** the method is equivalent to $h\|q\|_{L^\infty}/2$ and the stabilized scheme consists to find $\widehat{u}_k \in X_k^1(x_a, x_b)$ such that

$$\widehat{B}(\widehat{u}_k, v_k) = Lv_k, \ \forall v_k \in X_{k,0}^1(x_a, x_b)$$

where

$$\widehat{B}(u_k, v_k) = B(u_k, v_k) + \frac{h\|q\|_{L^\infty}}{2}\int_{x_a}^{x_b} u_k' v_k' dx.$$

The finite element method turn out to be unstable also for the approximation of reaction dominated problems. In this context, the **mass lumping technique** is often used to cure the instabilities. For linear finite elements, it corresponds to modify the nodal equations of the scheme, namely replacing $(u_{i+1} + 4u_i + u_{i-1})/6$ with the single value u_i. The error introduced by this modification can be thoroughly analyzed and, in conclusion, it does not substantially affect the convergence of the scheme[6].

8.8.4 The finite element method in two space dimensions

First, we address the problem of partitioning two dimensional domains into elements. The simplest case, consists in subdividing a polygonal domain $\Omega \subset \mathbb{R}^2$ into triangles. However, not all possible partitions are admissible for the construction of a finite element space. The following requirements must be satisfied by any admissible **triangulation**:

a) all elements K of the partition are triangles. We assume K is an open set;
b) for any distinct K_1, K_2 we have $K_1 \cap K_2 = \emptyset$;
c) if $\overline{K}_1 \cap \overline{K}_2 = E \neq \emptyset$, then E is either an entire face or a vertex;
d) the diameter of each element K, i.e. $h_K = \mathrm{diam}(K)$, is upper bounded, namely there exists $h > 0$ such that $h_k < h$.

The value h is called the characteristic mesh size. Furthermore, a triangulation is called quasi-uniform when it satisfies the following additional constraints:

e) there exists a constant $C > 1$ such that for any K the radii of the inner circle and outer circle with respect to K, i.e. ρ_K and h_K respectively, satisfy $1 < h_K/\rho_K < C$;
f) there exist two constants $C, c > 0$ such that for any K it holds $ch < h_K < Ch$.

[6] For further details, we refer to Section 5.5, Chapter 5, *Quarteroni* (2009).

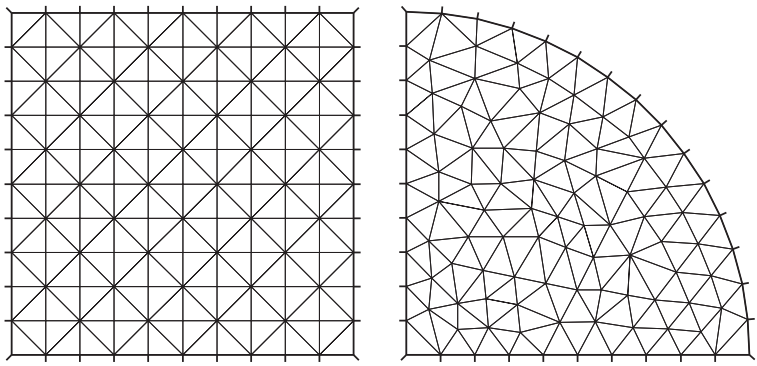

Fig. 8.4. A uniform (left) and quasi-uniform (right) partitions. We see that the finite element method easily adapts to generic configurations of the domain

The collection of all triangles of the partition is usually denoted as T_h. In Fig. 8.4 we show an example of uniform partition on the left and quasi uniform partition on the right.

To build up the finite element method in two space dimensions, we denote by $\mathbb{P}^r(K)$ the space of polynomial functions with maximum degree less or equal to r over the triangle K. Then the finite element space of degree r over the mesh T_h is given by

$$X_k^r = \{v_k \in C^0(\Omega) \ : \ v_k|_K \in \mathbb{P}^r(K), \ \forall K \in T_h\}$$

whose dimension is denoted as $N_k^r = \dim(X_k^r)$. As an instance, for linear finite elements in two space dimensions N_k^1 is equivalent to the number of vertexes of the computational mesh. This conclusion naturally emerges when looking at the construction of a Lagrangian basis for the linear finite element space. Reasoning as for the one dimensional case, the Lagrangian basis is given by the collection of functions that vanish in all but one vertex of the mesh. A component of such basis is depicted in Fig. 8.5 and according to their shape, these are often called *hat functions*.

The fundamental properties of the finite element method that we have presented for the approximation of one dimensional problems also apply to multiple space dimensions. In particular Theorem 8.7 and Corollary 8.1 still hold true, but particular care should be devoted to check the regularity requirement for the exact solution to be approximated, namely $u \in H^s(\Omega)$ with $s \geq 2$. Indeed, it is sufficient to consider a second order problem over a non convex polygonal domain to violate the previous regularity condition[7]. In the

[7] See *Salsa* (2004), Chapter 9.

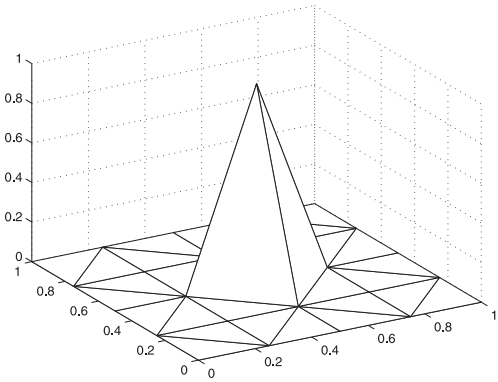

Fig. 8.5. A Lagrangian basis function of a two dimensional linear finite element space

forthcoming section, we will investigate the behaviour of the finite element method in these sub-optimal regularity conditions[8].

8.9 Exercises

8.1. Consider the following boundary value problem

$$\begin{cases} -u'' = 5x - 1 & \text{in } (-1, 2) \\ u(-1) = 1/2 \\ u'(2) = 2. \end{cases} \tag{8.84}$$

Write the associated variational formulation and determine existence, uniqueness and stability of the solution.

8.2. Analyze the following boundary value problem

$$\begin{cases} -(\mu(x)u')' + (\beta(x)u)' + \sigma(x)u = f(x) & a < x < b \\ u(a) = 0, \quad \mu(b)u'(b) = \beta(b)u(b) \end{cases} \tag{8.85}$$

where $a, b \in \mathbb{R}$ and μ, β, σ are given functions, with $\mu \geq \mu_0 > 0$. Introduce the functional spaces in order to write the variational formulation of the problem and prove that it is well-posed, under suitable assumptions on the data.

8.3. Consider the function

$$f(x) = \begin{cases} 2\pi^2 \sin \pi x & 0 \leq x < \dfrac{1}{2} \\ 2\pi^2 \sin \pi x - 6\left(\dfrac{1}{2} - x\right) + \left(\dfrac{1}{2} - x\right)^3 & \dfrac{1}{2} \leq x \leq 1. \end{cases}$$

[8] For a detailed analysis of the finite element method we refer the interested reader to *Quarteroni, Valli* (1997).